

Probability through ticket triage

Lecture 01 · Live writing edition

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A ticket-routing review

A service desk checks whether an LLM routes tickets to the appropriate teams. A reviewer supplies reference decisions; the model supplies classifications. What would it mean for them to agree?

Six reviewed tickets

For this hypothetical audit, the model uses a 0.50 threshold separately on each destination.

Ticket	Request	Reviewer	Model
t_1	Duplicate charge; cannot log in	Billing + Account Support	Billing
t_2	Duplicate monthly charge	Billing	Billing
t_3	MFA phone change; cannot log in	Account Support	Account Support
t_4	Delivery-status question	Other	Account Support
t_5	Duplicate renewal charge	Billing	Billing
t_6	Locked after password reset	Account Support	Account Support

What does the table count as an agreement?

Which rows agree?

Mark exact agreement between the reviewer reference and model classification. How many of the six records agree?

What does four out of six describe?

The table records four agreements. Is 4/6 a probability for every question about this model?

Equal-likelihood counting

If one of the six ticket IDs is selected uniformly, what is the probability that the selected record shows agreement?

The cases being counted matter

The occupied reviewer/model pairs are: both destinations / Billing (t_1); Billing only / Billing (t_2, t_5); Account only / Account (t_3, t_6); Other / Account (t_4). If one pair type were selected uniformly, what agreement probability would result?

A larger set of possible label pairs

There are four possible reviewer reference types and two model classes. How many pairs are possible in their full Cartesian product? How many appear here?

Frequency and repeated observations

Does an agreement fraction of $4/6$ establish a long-run agreement rate for future tickets?

Belief and evidence

Suppose an analyst assigns probability 0.8 to agreement on the next ticket, given this audit and other information. Does the table alone determine 0.8?

What the earlier views contribute

What does each account emphasize: equal-likelihood counting, frequency, belief, or evidential support? What remains necessary for a calculation?

A second question: a random chord

A chord is selected at random from a circle. What is the probability that its length exceeds the side of an inscribed equilateral triangle?

What must a probability model specify?

The ticket audit and the chord question both involve uncertainty. What mathematical objects would let us state the outcomes, the event and its probability?

Specify the ticket experiment

The six records are fixed. We select one ticket ID uniformly. What varies, and what are the possible outcomes?

Build the Billing event

Which ticket IDs have Billing in the reviewer's reference outcome? Write them as a set B .

Build the Account Support event

Which IDs have Account Support in the reviewer's reference outcome? Write this event as A .

Other as a complement

In this two-destination question, “Other” means neither Billing nor Account Support is required. Express this event using A and B .

Union and intersection

Use the ticket IDs to find $B \cap A$ and $B \cup A$. Interpret each event in words.

Count the union without double-counting

Why is the number of IDs in $B \cup A$ not $|B|+|A|$? Calculate the probability under uniform selection.

Pause and check the events

Without looking back, list the IDs in B , A , $B \cap A$, $B \cup A$ and O . Explain the overlap.

The probability-space triplet

For uniform selection from the six ticket IDs, specify the sample space, the events and the probability assignment.

Outcomes, events and probability values

Classify each object: t_2 , B , and $P(B) = 1/2$. Which one is the outcome? Which is an event? Which is a number?

Requirements on the event collection

If E is an event, should “not E ” also be an event? If $E_1, E_2, \dots$ are events, what about “at least one occurs”?

Kolmogorov's probability axioms

What conditions should a probability assignment satisfy on events?

A large grid of dots for taking notes, consisting of 20 rows and 100 columns of small, light gray dots.

A random variable for Billing

Can we record numerically whether Billing is required? Define X_B on the ticket outcomes.

A large grid of dots for taking notes.

Image of a set of outcomes

For $A = \{t_1, t_3, t_6\}$, what values does X_B produce? Write the image $X_B(A)$.

Inverse image and probability

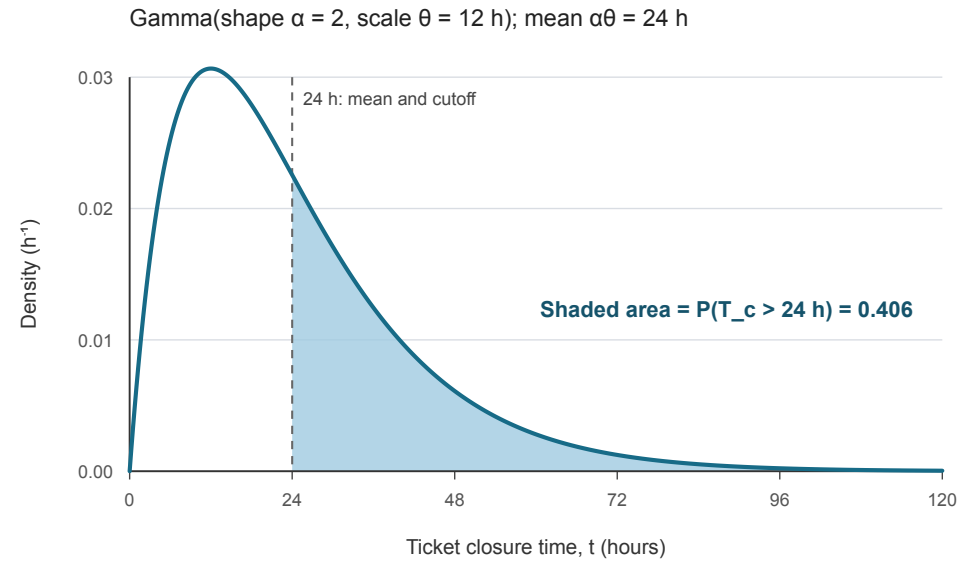
Which outcomes produce the value 1? Express them as $X_B^{-1}(\{1\})$ and calculate $P(X_B = 1)$.

A continuous ticket-time question

Let T_c be the closure time, in hours, of a ticket from a specified process. Under an illustrative Gamma model with shape 2 and scale 12 hours, calculate $P(T_c > 24)$.

From counting to integration

For the six equally likely ticket IDs, an event probability is found by counting IDs. For continuous closure time under this density, what replaces counting individual durations?



Evaluate the Gamma tail

Evaluate $\int_{24}^{\infty} \frac{t}{144} e^{-t/12} dt$ using $u = t/12$. State the interpretation.

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The closure-time probability space

State the triplet for the closure-time calculation and define its random variable explicitly.

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Worked check: neither named destination

Under uniform selection from the six audited IDs, find the probability of “Other,” meaning neither Billing nor Account Support is required.

Worked check: exactly one destination

Find the event and probability that exactly one of Billing and Account Support is required.

A large grid of dots for working out the solution.

Worked check: exact agreement

Let C be the event that the model classification equals the reviewer reference outcome. Find C and its probability in the uniform ticket-ID experiment.

A large grid of dots for writing the answer.

Worked check: agreement as a random variable

Define $Y(t) = 1$ for exact agreement and 0 otherwise. Find $Y^{-1}(\{1\})$ and $P(Y = 1)$.

A large grid of dots for working space, consisting of 20 rows and 100 columns of small, light blue dots.

Stopping point and continuation

State one distinction from today that is needed before interpreting an AI evaluation score as a future probability.